QSchView Help

Version 0.1.0

About QSchView

QSchView shows, prints, and exports QSpice schematics (`.qsch` files). The QSchView Project is a sub-project of The QParser2 Project. QSchView is a companion to QSchEdit, that project's schematic editor, but it does not edit anything: It never changes or saves the schematic it shows.

QSchView can:

- Show one schematic per window, with zoom and pan, and open the schematic behind a hierarchical block with a double-click.
- Print it fitted to one page or at a chosen scale across several pages with a live preview and with headers/footers.
- Write it to a PDF file.
- Export it to a layered SVG file ready for Inkscape or another SVG editor.
- Do all of the above from a command line, for batch work.

QSchView does not simulate, netlist, or edit. Symbol files (`.qsym`) are not supported yet; opening one says so.

Requirements. Windows 10 or later, 64-bit. QSpice does not have to be installed. When it is, QSchView draws in the colors QSpice is set to use; otherwise it uses QSpice's standard colors (see *Colors*).

Installing QSchView

The QSchView Project is a sub-project of The **QParser2** Project in the repository at https://github.com/robdunn4/QSpice. Compiled ready-to-run files are in the project's `binaries` folder.

To install:

1. Make an empty folder for QSchView somewhere you can write to, such as `C:\Tools\QSchView`. Avoid `C:\Program Files`: QSchView keeps its settings (and, when enabled, its log) beside the program (see *Settings and log files*) so the folder must be writable.
2. Download QSchView_v0.1.0.zip from the repository project `binaries` folder into your folder. Extract the files keeping any sub-folders as they are. Besides `QSchView.exe` and `QSchView.com` there are Qt libraries which QSchView needs; it will not start without them.

3. Run `QSchView.exe`. Nothing else needs installing.

Optional:

- **Open schematics from Explorer:** Right-click a `.qsch` file, choose *Open with* > *Choose another app*, and pick `QSchView.exe`.
- **Use the command line from any folder:** Add the QSchView folder to the Windows `PATH` (see *Command line*).
- **Help:** Help > QSchView Help opens `QSchView_Help.pdf`, if it is in the same folder as `QSchView.exe`.

To remove QSchView, simply delete the installation folder. QSchView does not modify the Windows Registry so there's nothing to uninstall.

Building from source. QSchView's sources are in the `sources` folder beside `binaries`, in the same repository. Building them needs Microsoft Visual Studio 2026 and Qt 6 (QSchView is built with Qt 6.11.2, MSVC, 64-bit).

Opening Schematics

There are several ways to open a schematic:

- **File > Open** (Ctrl+O). Several files can be selected at once.
- **File > Open Recent**, the most recent files: 20 by default, set by `MaxRecentFiles` in `QSchView.ini` (see *QSchView.ini*). The list is shared by every QSchView window. A file that no longer exists can be removed from the list when you choose it.
- **Drag and drop** one or more `.qsch` files from Explorer onto the window.
- **From Explorer**, with *Open with* or a file association.
- **From a command line:** `QSchView Amp.qsch`.

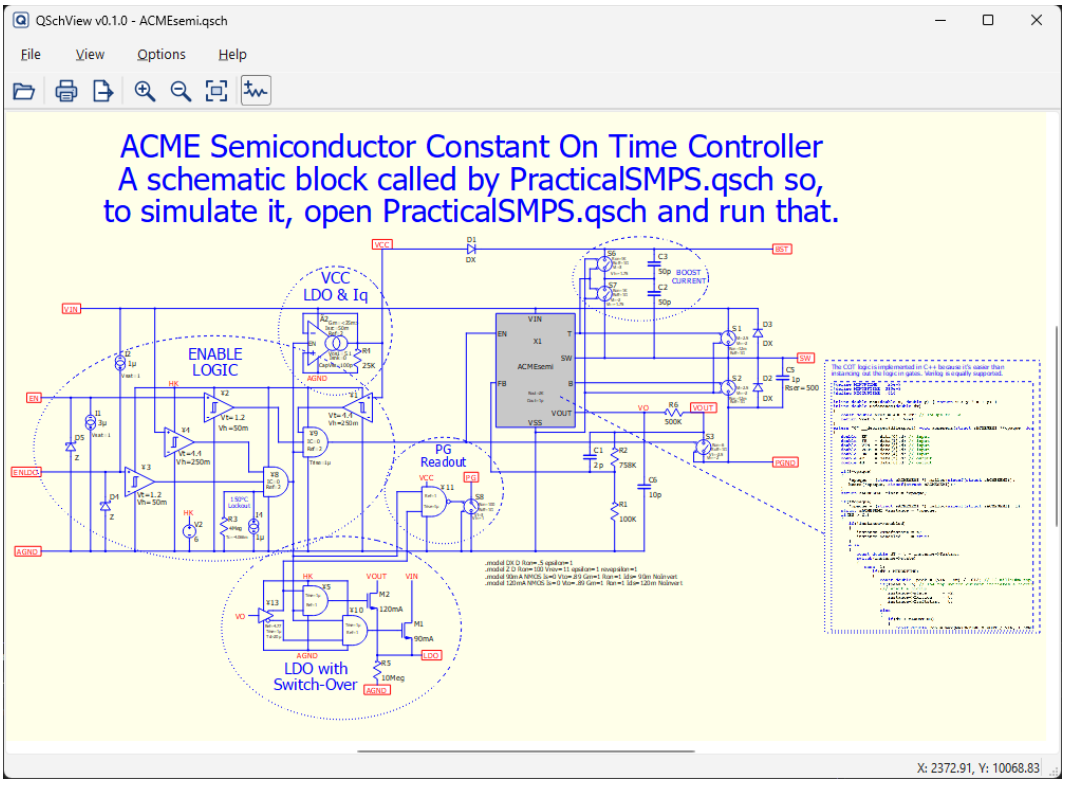
One schematic per window. Each window shows one schematic for as long as it is open. If the window is still empty, it shows the first file opened; every other file opens in a window of its own. To compare two schematics, open both and arrange the windows side by side.

Hierarchical blocks. Double-click a hierarchical block to open the schematic it links to in a window of its own. If the schematic cannot be found, a message says why and names the missing file.

Malformed schematic files. A properly formed schematic is shown complete. A malformed one (damaged, or written by another program) can contain records that are not allowed where they are, for example a wire inside a symbol. QSchView leaves those records out and says so in the status bar. When logging is enabled, the log identifies them (see *Settings and log files*).

The Main Window

QSchView starts with an empty window. Open a schematic with **File > Open** (Ctrl+O) or drag a `.qsch` file onto the window (see *Opening schematics*) and the window shows it:



The main window showing a schematic, with the toolbar.

The toolbar holds the main commands: Open | Print, Export as SVG | Zoom In, Zoom Out, Zoom to Fit | Show Resistor Polarity. Hold the pointer over a button to see its name and shortcut.

Command	Menu	Shortcut
Open	File	Ctrl+O
Open Recent	File	
Print	File	Ctrl+P
Export as SVG	File	
Exit	File	
Zoom In	View	Ctrl++
Zoom Out	View	Ctrl+–
Zoom to Fit	View	Ctrl+0
Show Resistor Polarity	Options	
Select Color Theme	Options	
Printing Defaults	Options	
Headers/Footers	Options	
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Moving around the drawing:

- **Mouse wheel:** Zooms in or out toward the pointer.

- **Middle button:** Hold it down and drag to pan.
- **Zoom In / Zoom Out:** One step, about the middle of the window.
- **Zoom to Fit:** Shows the whole schematic. A newly opened schematic starts this way.

Zoom stops at a minimum and maximum (see *Settings and log files*).

Status bar. The pointer's position is shown in schematic units at the bottom right.

Options

The **Options** menu holds QSchView's preferences. Each is saved in `QSchView.ini` as soon as it is changed and works with or without a schematic open.

- **Show Resistor Polarity**: When enabled, displays a small "**+**" beside each resistor's first pin so you can see which way round it was placed. (QSpice does not draw this.) It shows on screen, in print, in PDF, and in SVG exports.
- **Select Color Theme…:** Not available yet. Colors come from QSpice's settings or from `QSchView.ini` (see *Colors*).
- **Printing Defaults…:** What the Print dialog opens with (see *Printing defaults*).
- **Headers/Footers…:** The default header and footer text (see *Header and footer*). The sample shows the text for the open schematic or an example when none is open.

Colors

Eventually, QSchView will support color themes (like the QColorPrefs project). For now, it takes colors from (in order):

1. QSpice's own settings when QSpice is installed;
2. The `[Colors]` section of `QSchView.ini` for anything QSpice does not supply. On a computer without QSpice, that is everything;
3. The values of a standard QSpice installation.

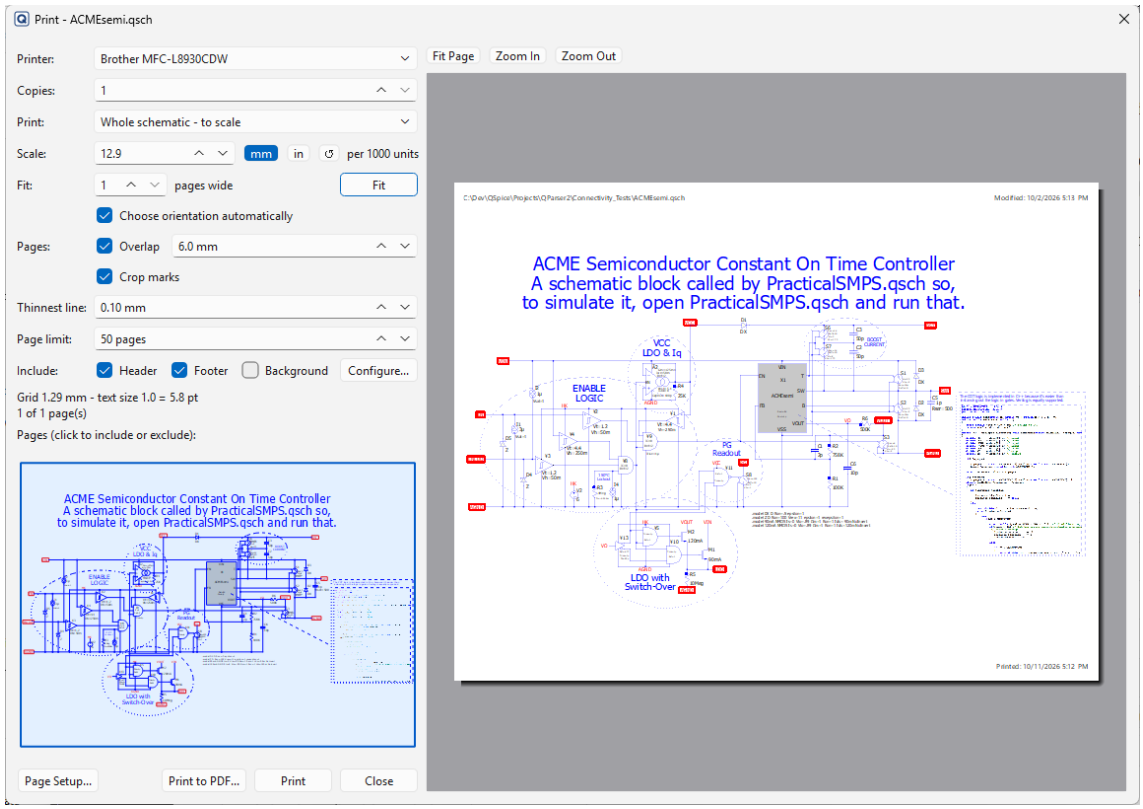
So on a computer with QSpice, the schematic looks as it does in QSpice. The `[Colors]` section matters only where QSpice is not installed, or does not have a setting.

Colors in `QSchView.ini` are written as QSpice writes them: `0xBBGGRR`, blue first. For example, `CADwireColor=0x00A000` makes wires green.

Printing

File > Print (Ctrl+P) opens the Print dialog. It is the only way to print: Everything that changes the output is set here. The preview always shows exactly what will print and **Print** prints it after one last confirmation (no Windows printer dialog follows).

Changes are for this print only. The Print dialog always opens with the printing defaults (see *Printing defaults*). Whatever you change in it (settings, printer, Page Setup) applies to this print and is forgotten afterwards. The one exception is the header and footer text which you can choose to keep (see *Header and footer*).



The Print dialog after Fit: The whole schematic on one page at 12.9 mm per 1000 units.

Before printing: Has the file changed?

Before printing or exporting, QSchView checks whether the schematic's file has changed since it was opened (for example, saved again from QSpice or QSchEdit). If it has, it asks:

- **Yes:** Reload the file first, keeping the current zoom and position;
- **No:** Print what is shown;
- **Cancel:** Do nothing.

If the file can no longer be read (moved or deleted), it asks whether to go on with what is shown.

Printer and copies

Printer lists the printers installed in Windows. **Copies** sets how many copies to print.

If Windows cannot open the chosen printer (it is off, asleep, or not connected), a dialog says so instead of showing a preview. Wake the printer, then choose another printer and back again, or close the dialog and open it again. A network printer that is asleep may first show Windows' own "connecting to printer" message: Wait for it rather than canceling it.

What to print

Print chooses what is printed and how:

Choice	**Prints**
Visible area - fit to page	What the window shows, fitted to one page.
Whole schematic - fit to page	The whole schematic, fitted to one page.
Whole schematic - to scale	The whole schematic at the scale below, on as many pages as it needs.
Visible area - to scale	What the window shows at the scale below.

The default is *Whole schematic - to scale*.

Scale

The **scale** is the length on paper of 1000 schematic units. The default is 25.4 mm (1 inch) per 1000 units. It can be shown in millimeters (to 0.1 mm, from 2.5 to 254 mm) or inches (to 0.01 inch, from 0.10 to 10 inches); click **mm** or **in** to switch. The reset button returns to the default.

Fit calculates the scale that makes the drawing a chosen number of **pages wide** (rounded down).

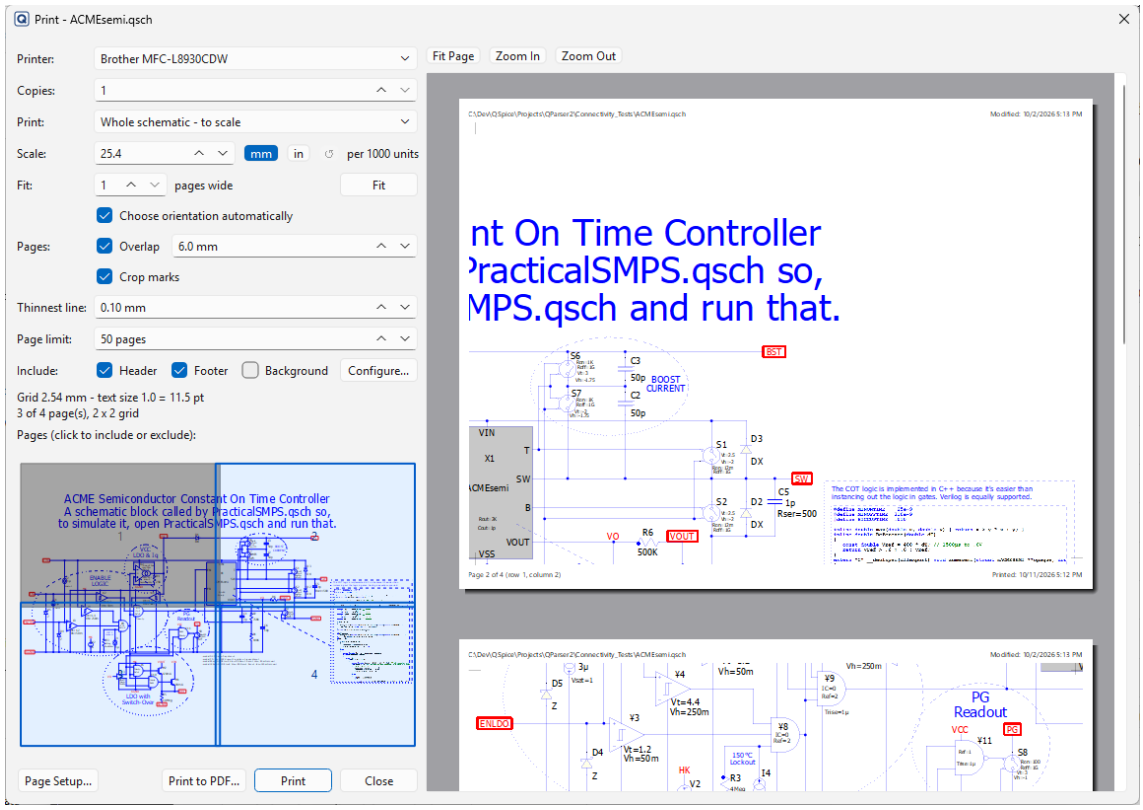
Below the settings, the dialog shows what the scale means on paper (the size of QSpice's 100-unit grid, and the point size of normal text) and how many pages will print.

Several pages

At scale, a large schematic is printed on a grid of pages. It optionally includes markers to simplify cutting/joining pages into a single large print-out:

- **Overlap:** The strip, in mm, that neighboring pages share (6 mm by default), so they can be trimmed and joined.
- **Crop marks:** Marks on each page where it meets its neighbors.
- **Page limit:** The maximum number of pages one print may use. A job that would need more is not printed in part: The dialog asks for a smaller scale or a higher limit.

The page map shows the area being printed with the page grid over it. Click a page to leave it out and again to put it back. Pages with nothing on them are skipped.



The Print dialog at scale on four pages with page 1 left out in the page map.

Other settings

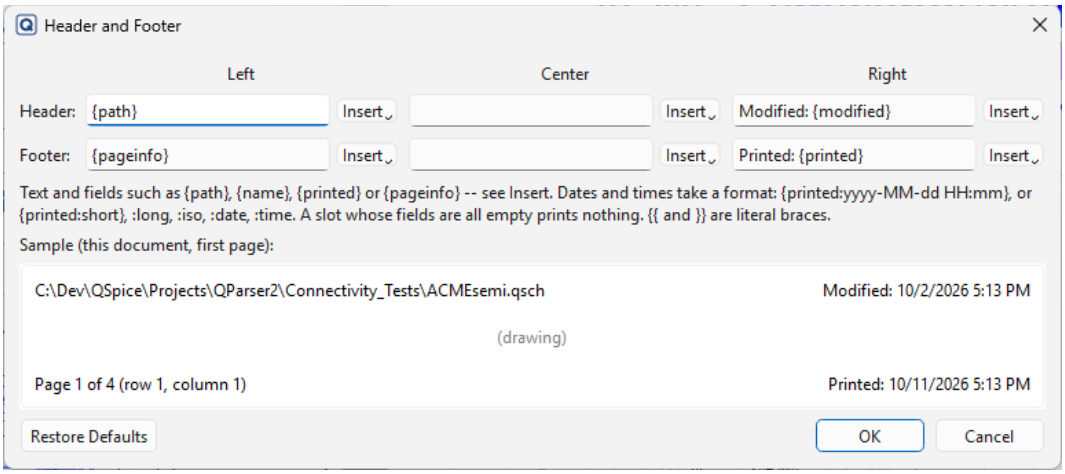
- **Choose orientation automatically**: Turns the page to whichever orientation needs fewer pages (or, when fitting, gives the larger drawing).
- **Thinnest line:** No line prints thinner than this (0.10 mm by default) so fine lines do not vanish when printed small.

Header and footer

Include turns the **Header** and **Footer** on or off. A band that is off gives its space to the drawing. **Configure…** sets what they say. **Background** prints the canvas color behind the drawing.

The same dialog opens from **Options > Headers/Footers…**, where it changes the default text. From the Print dialog's **Configure…**, if you change the text, QSchView asks whether to make it the default:

- **Yes:** Use it for this print and keep it as the default;
- **No:** Use it for this print only;
- **Cancel:** Go back to editing.



The Header and Footer dialog.

Each band has three slots: **Left**, **Center**, and **Right**. A slot holds ordinary text and **fields**, names in braces that are replaced when each page prints. **Insert** adds a field at the cursor; the **Sample** shows the result for the current schematic as you type.

Field	Becomes	Example
`{path}`	The schematic's full path.	`C:\Dev\Amp\Amp.qsch`
`{name}`	The file name only.	`Amp.qsch`
`{folder}`	The folder only.	`C:\Dev\Amp`
`{title}`	The schematic's name.	`Amp.qsch`
`{modified}`	The file's last-modified date and time.	`10/5/2026 3:42 PM`
`{printed}`	The date and time of printing.	`10/5/2026 3:42 PM`
`{page}`	This page's number.	`2`
`{pages}`	The number of pages printed.	`6`
`{row}`, `{col}`	This page's row and column in the page grid.	`1`, `2`
`{pageinfo}`	`Page 2 of 6`, plus the row and column when tiled; empty for one page.	`Page 2 of 6 (row 1, column 2)`
`{printer}`	The printer's name or `PDF file`.	`HP LaserJet M404`
`{scale}`	The scale the page is printed at.	`25.4 mm (1.00 inch) per 1000 units`
`{user}`	Your Windows user name.	`Robert`

Field names are not case-sensitive.

Dates and times. `{modified}` and `{printed}` take a format after a colon. With none, they use Windows' short date and time.

Format	Gives	Example
(none) or `short`	Windows' short date and time.	`10/5/2026 3:42 PM`
`long`	Windows' long date and time.	`Monday, October 5, 2026 3:42:07 PM`
`iso`	Year-month-day, 24-hour time.	`2026-10-05 15:42:07`
`date`	The short date only.	`10/5/2026`

Format	Gives	Example
`time`	The short time only.	`3:42 PM`

Anything else is a pattern built from these codes:

Code	Means	Code	Means
`yyyy` / `yy`	Year, 4 or 2 digits.	`H` / `HH`	Hour, 24-hour.
`M` / `MM`	Month number.	`h` / `hh`	Hour, 12-hour (with `AP`).
`MMM` / `MMMM`	Month, short or full name.	`AP` / `ap`	AM/PM.
`d` / `dd`	Day.	`m` / `mm`	Minute.
`ddd` / `dddd`	Weekday, short or full name.	`s` / `ss`	Second.
`'text'`	Literal text.	`t`	Time zone.

For example, `{printed:yyyy-MM-dd HH:mm}` prints `2026-10-05 15:42`, and `{modified:d MMM yyyy 'at' HH:mm}` prints `5 Oct 2026 at 15:42`.

Rules:

- A slot whose fields all come out empty prints nothing, its own text included. So `{pageinfo}` prints nothing on a single page.
- An unknown field prints as typed, braces and all, so a typo is easy to spot.
- `{{` and `}}` print a single `{` and `}`.
- The Center slot stays centered. Text too long for its space is shortened in the middle, so its start and end stay readable.

Restore Defaults puts back: Header `{path}` and `Modified: {modified}`; footer `{pageinfo}` and `Printed: {printed}`.

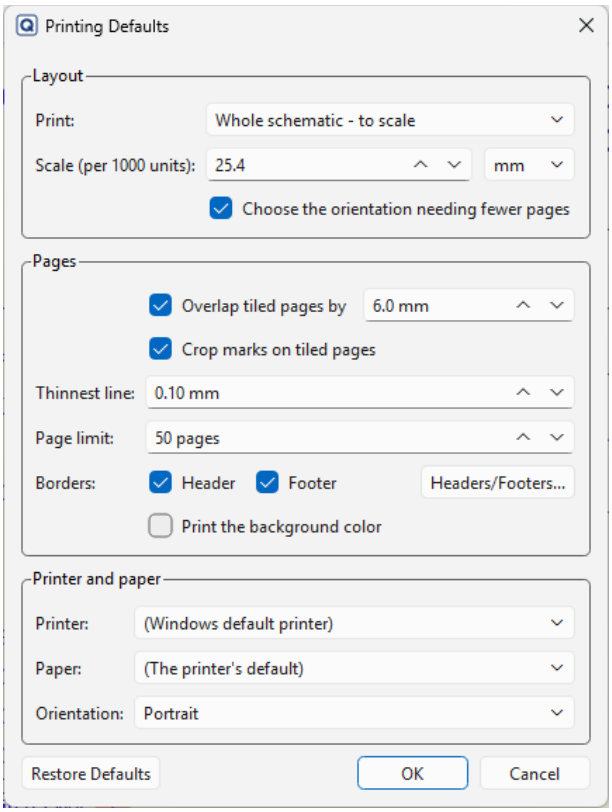
Page Setup and Print to PDF

Page Setup… chooses the paper, orientation, and margins, and gives access to the printer's own properties. It changes settings only; it does not print.

Print to PDF… writes the same pages, exactly as previewed, to a PDF file. It suggests the schematic's own name and folder.

Printing defaults

Options > Printing Defaults… sets what the Print dialog opens with. It is the only place (with Options > Headers/Footers) where print settings are saved. They are kept in `QSchView.ini` and used by every QSchView window and by the command line tool.



The Printing Defaults dialog.

Setting	**Default**
Print	Whole schematic - to scale
Scale	25.4 mm (1 inch) per 1000 units, shown in mm
Choose the orientation needing fewer pages	On
Overlap tiled pages by	On, 6 mm
Crop marks on tiled pages	On
Thinnest line	0.10 mm
Page limit	50 pages
Header, Footer	On
Print the background color	Off
Printer	The Windows default printer
Paper	The printer's default
Orientation	Portrait

- **Print the background color:** Fills each page with the canvas color, as on screen. Off, the paper shows through, which saves ink.
- **Headers/Footers…** edits the default header and footer text. Saved with the rest when you click OK.

- **Printer and paper** are chosen by name without contacting the printer so a printer that is off or asleep does not hold the dialog up. A saved printer that is no longer installed is shown as *not installed*.
- **Restore Defaults** puts back the values in the table; nothing is saved until you click OK.

Exporting to SVG

File > Export as SVG writes the whole schematic to an SVG file. It suggests the schematic's own name and folder.

- **Size:** Always 25.4 mm (1 inch) per 1000 units, at true size in millimeters, so it opens at real size in Inkscape. The size can be changed in the SVG editor without loss: Everything is drawn as lines and shapes, not pixels.
- **Line width:** No line is thinner than `SvgMinLineWidthMm` in the `[Export]` section of `QSchView.ini` (0.1 mm by default). Scaling the drawing down in an editor thins its lines in proportion, unless the editor is set not to scale line widths.
- **No print settings apply:** No header or footer, on a single page, and the background is always included (on its own layer, which can be hidden).

Layers. The file is built for editing. Each kind of element is on a layer of its own (Inkscape's layers), from bottom to top:

Layer	Holds
Background	One rectangle in the canvas color; locked.
Graphics	Lines, rectangles, ellipses, arcs, and images drawn on the schematic.
Symbols	One group per component, named by its designator (`R1`, `U3`).
Wires	One sub-group per net, holding that net's wires and bus taps.
Net labels/ports	One group per label or port.
Junctions	One group per junction dot.
Text	Free text, comments, and directives.

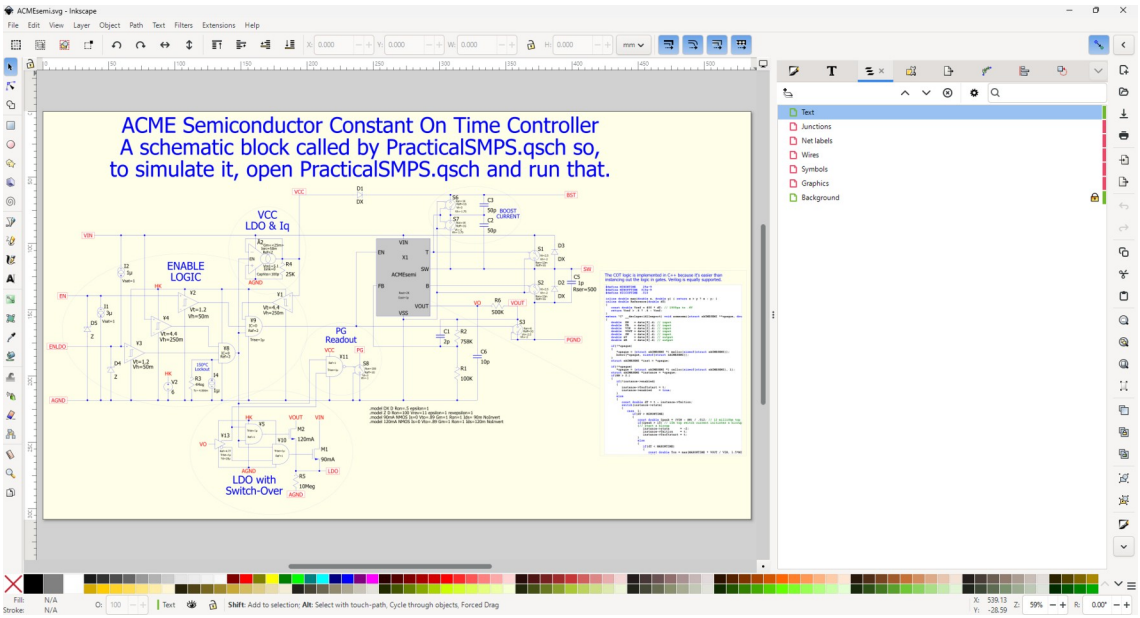
Within each layer, elements keep their stacking order from the schematic.

Information carried in the file:

- Each component's group has its designator as its ID and label, a title (shown as a tooltip by many viewers), and the attributes `data-refdes`, `data-value`, and `data-symbol`.
- Net labels, ports, and junctions carry their net's name as `data-net`.

In Inkscape:

- Open the Layers panel (Layer > Layers and Objects) to hide, show, or lock whole kinds of elements: For example, hide the background for a transparent drawing.
- Each net's wires are one group: Select the group to recolor or thicken a whole net.
- Edit > XML Editor shows each component's designator, value, and symbol.



An exported SVG in Inkscape with the Layers panel open.

Command line

QSchView can print and export without a window for batch work. It handles **one schematic per run**, prints what it did, and returns a result code.

QSchView.com and QSchView.exe. QSchView is installed as two files side by side. `QSchView.exe` is the program. `QSchView.com` is a small console launcher: At a command prompt, typing `QSchView` runs it (Windows tries `.com` before `.exe`). It starts `QSchView.exe`, waits for it, shows its messages, and passes back its result code, so batch files and scripts work as expected. Without options, it simply opens the GUI window. To use `QSchView` from any folder, add the QSchView folder to the Windows `PATH`; otherwise give its full path.

Usage

```
QSchView FILE.qsch [options]      print and/or export one schematic
QSchView --list-printers          list the installed printers
QSchView --help | --version
```

Output (at least one):

Option	Does
`--print`	Prints on the saved printer or the Windows default.
`--print=PRINTER`	Prints on the named printer (see `--list-printers`).
`--pdf=FILE`	Writes a PDF; needs no printer.
`--svg=FILE`	Writes an SVG; no print settings apply (see *Exporting to SVG*).

Page (otherwise taken from the printing defaults):

Option	Does
`--fit`	The whole schematic fitted to one page.
`--scale=MM`	At scale: MM on paper per 1000 units on as many pages as needed.
`--orientation=O`	`portrait`, `landscape`, or `auto`
`--paper=P`	`A4`, `Letter`, `Legal`, `A3`, `Tabloid`, …
`--header`, `--no-header`	Header on or off.
`--footer`, `--no-footer`	Footer on or off.
`--background`, `--no-background`	The canvas color behind the drawing on or off (print and PDF).
`--copies=N`	Printed copies.

A value may follow its option after = or as the next argument (`--pdf out.pdf`), except a printer name which needs =.

There is no window, so "visible area" does not apply: The whole schematic is printed. Command-line choices are used for that run only; they never change the printing defaults.

Result codes:

Code	Means
0	Done.
1	Bad arguments.
2	The file cannot be used: Not found, not a schematic, or a `.qsym`.
3	The printer is not installed or Windows could not open it.
4	Output failed or the job needs more pages than the page limit.

When several outputs are asked for, each is tried even if an earlier one fails; the result is the first failure.

Examples

```
QSchView Amp.qsch --pdf=Amp.pdf
QSchView Amp.qsch --print="HP LaserJet M404" --scale=10 --paper=A3
QSchView Amp.qsch --svg=Amp.svg --pdf=Amp.pdf --no-header
```

A batch file that writes a PDF of every schematic in a folder:

```
@echo off
for %%f in (*.qsch) do (
  QSchView "%%f" --fit --pdf="%%~nf.pdf"
  if errorlevel 1 echo Failed: %%f
)
```

Settings and Log Files

QSchView.ini

QSchView keeps its settings in `QSchView.ini` beside `QSchView.exe`. It is written the first time QSchView runs with every setting present so each can be found and changed by hand:

```
MinScale=[] // default: 0.02
MaxScale=40 // default: 20
```

`[]` means "the default"; any other value overrides it. Close every QSchView window before editing the file.

Section	Holds
`[QSpice]`	`QInstallDirKey`: Where in the registry QSpice keeps its settings. Leave it unless QSpice is installed unusually.
`[View]`	`MinScale`, `MaxScale`: The zoom limits, in pixels per schematic unit. `NewViewWidth`: Schematic units across an empty view. `ShowResistorOrientation`: Show Resistor Polarity
`[Colors]`	Colors and sizes by QSpice's own names (see *Colors*)
`[Print]`	The printing defaults and the header/footer text. Written by Options > Printing Defaults and Headers/Footers.
`[Export]`	`SvgMinLineWidthMm`: The thinnest line in SVG exports.
`[Log]`	`MaxSizeKB`: The log file's size limit; 0 (the default) means no log file (see *QSchView.log*).
`[Files]`	`MaxRecentFiles`: How many files File > Open Recent lists, 1 to 99.
`[RecentFiles]`	The recently used files list.
`[Window]`	The window sizes and positions.

QSchView.log

QSchView can keep a log, `QSchView.log`, beside `QSchView.exe`: What was opened, printed, exported, records left out, and any problems. It is **off by default**, so nothing builds up beside the program. Turn it on when you need it, usually to report a problem:

1. Close every QSchView window.
2. In `QSchView.ini`, under `[Log]`, set `MaxSizeKB=1024` (any size in KB; 64 is the smallest used).
3. Start QSchView and do again whatever went wrong.
4. Send `QSchView.log` with your report (see *The Project*).
5. Set `MaxSizeKB=[]` again to turn the log off.

Every window adds to the same file; each line carries the time and the process number. When the file passes `MaxSizeKB`, it is renamed `QSchView.log.old` (replacing any older one) and a new log is started.

Troubleshooting

Problem	What to do
The Print dialog says Windows could not open the printer	The printer is off, asleep, or not connected. Wake it, then choose it again or reopen the dialog. Wait for Windows' "connecting to printer" message rather than canceling it.
"Nothing to print"	The schematic, or the visible area, is empty. Choose *Whole schematic*, or zoom to fit first.
The Print dialog forgot my changes	It always opens with the printing defaults. Set them with Options > Printing Defaults.
Double-clicking a block says the schematic was not found	The message names the file it looked for. Check the block's first attribute, and that the schematic is in the folder it names (or beside this schematic).
"Reload it first?" before every print	The file is being saved again by another program (QSchEdit, QSpice). Choose Yes to print the latest version.
A schematic will not open	The message gives the reason and the line. For the details, turn on the log (see *QSchView.log*) and open it again. `.qsym` files are not supported yet.
Something is missing from the drawing	If the file is malformed, the status bar says records were left out; with the log on, it lists them.
Colors differ from QSpice	Check `[Colors]` in `QSchView.ini`, and `QInstallDirKey` if QSpice is installed.
A batch run fails	Check the result code (see *Command line*) and the message; run it by hand to see it.

The Project

The QSchView Project is a sub-project of The QParser2 Project. Full sources, documentation, and the ready-to-run files (in the `binaries` folder; see *Installing QSchView*) are at **https://github.com/robdunn4/QSpice**.

Support. Please address questions and problem reports to the author, **@RDunn**, on Qorvo's QSpice support forum: **https://forum.qorvo.com/c/qspice/9**. Include the schematic, if you can, and `QSchView.log` from a run that shows the problem. The log is off by default; *QSchView.log* says how to turn it on.

Related programs: QSchEdit, the schematic editor. Both are built on the same shared libraries, listed with their versions in Help > About.

License: GNU General Public License version 3 (GPL-3.0).

Qt: QSchView uses the Qt framework, which has its own licenses. The Qt libraries supplied with QSchView are the unmodified open-source Qt 6.11.2 libraries, used under the GNU Lesser General Public License version 3 (LGPL-3.0). Qt sources and licensing details are available from The Qt Company; see also Qt Licensing.

QSpice is a product of Qorvo, Inc. QSchView is an independent project, not made or endorsed by Qorvo; questions about it go to its author, not to Qorvo.

Version history

Version	Date	Changes
0.1.0	*[date]*	First release